



Recitation 9

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1. Review

2. Examples

3. Recitation



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- Discrete case

$$p_Y(y) = \mathbf{P}(g(X) = y) = \sum_{x: g(x)=y} p_X(x)$$

- Continuous case

- Obtain CDF of Y : $F_Y(y) = \mathbf{P}(Y \leq y)$

- Differentiate to get $f_Y(y) = \frac{dF_Y}{dy}(y)$

- The linear case $Y = aX + b$

$$f_Y(y) = \frac{1}{a} f_X\left(\frac{y-b}{a}\right)$$

- The monotonic case

$$f_Y(y) = f_X(h(y)) \left| \frac{dh}{dy}(y) \right|$$

- Convolution

$$p_W(w) = \sum_x p_X(x)p_Y(w-x)$$
$$f_W(w) = \int_{-\infty}^{\infty} f_X(x)f_Y(w-x) dx$$

- Covariance and correlation

$$\text{cov}(X, Y) = \mathbf{E}[(X - \mathbf{E}[X]) \cdot (Y - \mathbf{E}[Y])] = \mathbf{E}[XY] - \mathbf{E}[X]\mathbf{E}[Y]$$

$$\rho = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1]$$

Independent $\implies \text{cov}(X, Y) = 0, \rho = 0$ (**converse is not true**)

- Variance of the sum of RVs

$$\text{var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{var}(X_i) + \sum_{\{(i,j)|i \neq j\}} \text{cov}(X_i, X_j)$$

Definition of Correlation Coefficient

The **correlation coefficient** $\rho(X, Y)$ of two RVs X and Y that have nonzero variance is defined as

$$\begin{aligned}\rho &= \mathbf{E} \left[\frac{(X - \mathbf{E}[X])}{\sigma_X} \cdot \frac{(Y - \mathbf{E}[Y])}{\sigma_Y} \right] \\ &= \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}\end{aligned}$$

- $-1 \leq \rho \leq 1$ (Proved by **Schwarz inequality**, see recitation)
- $|\rho| = 1 \Leftrightarrow (X - \mathbf{E}[X]) = c(Y - \mathbf{E}[Y])$ (linearly related)
- Independent $\Rightarrow \rho = 0$ (**converse is not true**)

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The difference of two independent r.v.'s

- Consider the case where X and Y are independent exponential RVs with parameter λ . Find the PDF for $X - Y$.



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- Consider the case where X and Y are independent exponential RVs with parameter λ . Find the PDF for $X - Y$.
- Fix some $z \geq 0$ and note that $f_Y(x - z)$ is nonzero when $x \geq z$. Thus,

$$\begin{aligned} f_{X-Y}(z) &= \int_{-\infty}^{\infty} f_X(x) f_{-Y}(z - x) dx \\ &= \int_{-\infty}^{\infty} f_X(x) f_Y(x - z) dx \\ &= \int_z^{\infty} \lambda e^{-\lambda x} \lambda e^{-\lambda(x-z)} dx \\ &= \frac{\lambda}{2} e^{-\lambda z}. \end{aligned}$$

- The answer for the case $z < 0$

$$f_{X-Y}(z) = f_{Y-X}(z) = f_{-(X-Y)}(z) = f_{X-Y}(-z)$$

- The first equality holds by symmetry.

The sum of two independent normal r.v.'s



- $X \sim N(\mu_x, \sigma_x^2)$, $Y \sim N(\mu_y, \sigma_y^2)$, independent
- Let $W = X + Y$, $W \sim N(\mu_x + \mu_y, \sigma_x^2 + \sigma_y^2)$
- We first prove the zero-mean case, and we can simply prove the general case by linearity property.





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- We first prove the zero-mean case, and we can simply prove the general case by linearity property.

$$\begin{aligned} f_W(w) &= \int_{-\infty}^{\infty} f_X(x) f_Y(w-x) dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma_x} \exp\left(-\frac{x^2}{2\sigma_x^2}\right) \frac{1}{\sqrt{2\pi}\sigma_y} \exp\left(-\frac{(w-x)^2}{2\sigma_y^2}\right) dx \\ &= \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{\sigma_y^2 x^2 + \sigma_x^2 (w-x)^2}{2\sigma_x^2\sigma_y^2}\right) dx \\ &= \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{(\sigma_x^2 + \sigma_y^2)x^2 - 2\sigma_x^2 wx + \sigma_x^2 w^2}{2\sigma_x^2\sigma_y^2}\right) dx \end{aligned}$$



(Cont'd)

$$\begin{aligned} &= \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{(\sigma_x^2 + \sigma_y^2)x^2 - 2\sigma_x^2wx + \sigma_x^2w^2}{2\sigma_x^2\sigma_y^2}\right) dx \\ &= \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{(\sigma_x^2 + \sigma_y^2)\left(x - \frac{\sigma_x^2w}{\sigma_x^2 + \sigma_y^2}\right)^2 + \sigma_x^2w^2 - \frac{(\sigma_x^2w)^2}{\sigma_x^2 + \sigma_y^2}}{2\sigma_x^2\sigma_y^2}\right) dx \\ &= \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{(\sigma_x^2 + \sigma_y^2)\left(x - \frac{\sigma_x^2w}{\sigma_x^2 + \sigma_y^2}\right)^2}{2\sigma_x^2\sigma_y^2}\right) \exp\left(-\frac{w^2}{2(\sigma_x^2 + \sigma_y^2)}\right) dx \end{aligned}$$



(Cont'd)

$$\begin{aligned} &= \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{(\sigma_x^2 + \sigma_y^2)(x - \frac{\sigma_x^2 w}{\sigma_x^2 + \sigma_y^2})^2}{2\sigma_x^2\sigma_y^2}\right) \exp\left(-\frac{w^2}{2(\sigma_x^2 + \sigma_y^2)}\right) dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi} \frac{\sigma_x\sigma_y}{\sqrt{\sigma_x^2 + \sigma_y^2}}} \exp\left(-\frac{(x - \frac{\sigma_x^2 w}{\sigma_x^2 + \sigma_y^2})^2}{2 \frac{\sigma_x^2\sigma_y^2}{\sigma_x^2 + \sigma_y^2}}\right) dx \\ &\quad \cdot \frac{1}{\sqrt{2\pi} \sqrt{\sigma_x^2 + \sigma_y^2}} \exp\left(-\frac{w^2}{2(\sigma_x^2 + \sigma_y^2)}\right) \\ &= \frac{1}{\sqrt{2\pi} \sqrt{\sigma_x^2 + \sigma_y^2}} \exp\left(-\frac{w^2}{2(\sigma_x^2 + \sigma_y^2)}\right) \sim N(0, \sigma_x^2 + \sigma_y^2) \end{aligned}$$

The sum of two independent normal r.v.'s



- For $X \sim N(\mu_x, \sigma_x^2)$, $Y \sim N(\mu_y, \sigma_y^2)$, consider $(X - \mu_x) \sim N(0, \sigma_x^2)$, $(Y - \mu_y) \sim N(0, \sigma_y^2)$
- We have $[(X - \mu_x) + (Y - \mu_y)] \sim N(0, \sigma_x^2 + \sigma_y^2)$
- Thus $(X + Y) \sim N(\mu_x + \mu_y, \sigma_x^2 + \sigma_y^2)$



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1. Let X have the normal distribution with mean 0 and variance 1. Also let $Y = g(X)$ where

$$g(t) = \begin{cases} -t, & \text{for } t \leq 0, \\ \sqrt{t}, & \text{otherwise.} \end{cases}$$

Find the probability density function of Y .



Solution:

$$\begin{aligned}F_Y(y) &= \mathbf{P}(Y \leq y) \quad \text{if } y \geq 0 \\&= \mathbf{P}(0 \leq \sqrt{X} \leq y) + \mathbf{P}(-y \leq X < 0) \\&= (F_X(y^2) - \frac{1}{2}) + (\frac{1}{2} - F_X(-y)) \\&= F_X(y^2) - F_X(-y)\end{aligned}$$

$$\begin{aligned}\Rightarrow f_Y(y) &= \frac{dF_Y}{dy}(y) \\&= 2yf_X(y^2) + f_X(-y) \\&= \frac{1}{\sqrt{2\pi}} \left(2ye^{-y^4/2} + e^{-y^2/2} \right), \text{ for all } y \geq 0\end{aligned}$$

2. Let X and Y be independent standard normal RVs. The pair (X, Y) can be described in polar coordinates in terms of RVs $R \geq 0$ and $\Theta \in [0, 2\pi]$, so that

$$X = R \cos \Theta, \quad Y = R \sin \Theta$$

Show that R and Θ are independent. Furthermore,

- (a) Find $f_R(r)$.
- (b) Find $f_\Theta(\theta)$.
- (c) Find $f_{R,\Theta}(r, \theta)$.

Solution:

(a)

$$\begin{aligned} F_R(r) &= \mathbf{P}(R \leq r) \\ &= \int_0^r \int_0^{2\pi} f_X(t \cos \theta) f_Y(t \sin \theta) d\theta dt \\ &= \int_0^r \int_0^{2\pi} \frac{1}{2\pi} e^{-t^2/2} d\theta dt \\ &= \int_0^{r^2/2} e^{-u} du \end{aligned}$$

$$\implies f_R(r) = \frac{dF_R}{dr}(r) = r e^{-r^2/2}, \quad r \geq 0$$

(b)

$$\begin{aligned}
 F_{\Theta}(\theta) &= \mathbf{P}(\Theta \leq \theta) \\
 &= \int_0^{\theta} \int_0^{\infty} f_X(r \cos t) f_Y(r \sin t) r dr dt \\
 &= \int_0^{\infty} r e^{-r^2/2} dr \int_0^{\theta} \frac{dt}{2\pi} \\
 &= \frac{\theta}{2\pi}, \quad 0 \leq \theta \leq 2\pi
 \end{aligned}$$

$$\Rightarrow f_{\Theta}(\theta) = \frac{1}{2\pi}, \quad 0 \leq \theta \leq 2\pi$$

(c)

$$\begin{aligned} F_{R,\Theta}(r, \theta) &= \mathbf{P}(R \leq r, \Theta \leq \theta) \\ &= \int_0^\theta \int_0^r \frac{1}{2\pi} e^{-r'^2/2} r' dr' d\theta' \\ &= \frac{\theta}{2\pi} (1 - e^{-r^2/2}), \quad r \geq 0, \quad 0 \leq \theta \leq 2\pi \end{aligned}$$

$$\begin{aligned} \Rightarrow f_{R,\Theta}(r, \theta) &= \frac{\partial}{\partial r} \frac{\partial}{\partial \theta} F_{R,\Theta}(r, \theta) \\ &= \frac{1}{2\pi} r e^{-r^2/2}, \quad r \geq 0, \quad 0 \leq \theta \leq 2\pi \end{aligned}$$

1. Show that

$$\rho(aX + b, Y) = \rho(X, Y).$$

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Proof:

$$\begin{aligned}\rho(aX + b, Y) &= \frac{\text{cov}(aX + b, Y)}{\sqrt{\text{var}(aX + b)\text{var}(Y)}} \\ &= \frac{a \cdot \text{cov}(X, Y)}{a\sqrt{\text{var}(X)\text{var}(Y)}} \\ &= \rho(X, Y)\end{aligned}$$

2. **Schwarz inequality.** Show that for any RVs X and Y , we have

$$(\mathbf{E}[XY])^2 \leq \mathbf{E}[X^2]\mathbf{E}[Y^2].$$

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Proof: If $\mathbf{E}[Y^2] = 0$, we have $Y = 0$ with probability 1 and $\mathbf{E}[XY] = 0$, under which the inequality holds. Otherwise, we have

$$\begin{aligned} 0 &\leq \mathbf{E}\left[\left(X - \frac{\mathbf{E}[XY]}{\mathbf{E}[Y^2]}Y\right)^2\right] \\ &= \mathbf{E}\left[X^2 - 2\frac{\mathbf{E}[XY]}{\mathbf{E}[Y^2]}XY + \frac{(\mathbf{E}[XY])^2}{(\mathbf{E}[Y^2])^2}Y^2\right] \\ &= \mathbf{E}[X^2] - 2\frac{\mathbf{E}[XY]}{\mathbf{E}[Y^2]}\mathbf{E}[XY] + \frac{(\mathbf{E}[XY])^2}{(\mathbf{E}[Y^2])^2}\mathbf{E}[Y^2] \\ &= \mathbf{E}[X^2] - \frac{(\mathbf{E}[XY])^2}{\mathbf{E}[Y^2]} \end{aligned}$$

3. Consider the correlation coefficient of two RVs X and Y that have positive variances. Show that:

(a) $|\rho(X, Y)| \leq 1$.

(b) If $Y - \mathbf{E}[Y]$ is a positive multiple of $X - \mathbf{E}[X]$, then $\rho(X, Y) = 1$.

(c) If $\rho(X, Y) = 1$, then, with probability 1, $Y - \mathbf{E}[Y]$ is a positive multiple of $X - \mathbf{E}[X]$.

Proof:

- (a) Let $\tilde{X} = X - \mathbf{E}[X]$ and $\tilde{Y} = Y - \mathbf{E}[Y]$. Using Schwartz inequality we have

$$(\rho(X, Y))^2 = \frac{(\text{cov}(X, Y))^2}{\text{var}(X)\text{var}(Y)} = \frac{(\mathbf{E}[\tilde{X}\tilde{Y}])^2}{\mathbf{E}[\tilde{X}^2]\mathbf{E}[\tilde{Y}^2]} \leq 1$$

and hence $|\rho(X, Y)| \leq 1$.

- (b) If $\tilde{Y} = \alpha\tilde{X}$, then

$$\rho(X, Y) = \frac{\mathbf{E}[\tilde{X}\tilde{Y}]}{\sqrt{\mathbf{E}[\tilde{X}^2]\mathbf{E}[\tilde{Y}^2]}} = \frac{\alpha\mathbf{E}[\tilde{X}^2]}{\alpha\mathbf{E}[\tilde{X}^2]} = 1.$$

(c) As $\rho(X, Y) = 1$, we have $\mathbf{E}[\tilde{X}\tilde{Y}] = \sqrt{\mathbf{E}[\tilde{X}^2]\mathbf{E}[\tilde{Y}^2]}$. Thus,

$$\mathbf{E}\left[\left(\tilde{X} - \frac{\mathbf{E}[\tilde{X}\tilde{Y}]}{\mathbf{E}[\tilde{Y}^2]}\tilde{Y}\right)^2\right] = \mathbf{E}[\tilde{X}^2] - \frac{(\mathbf{E}[\tilde{X}\tilde{Y}])^2}{\mathbf{E}[\tilde{Y}^2]} = 0.$$

It can be concluded that with probability 1,

$$\tilde{X} = \frac{\mathbf{E}[\tilde{X}\tilde{Y}]}{\mathbf{E}[\tilde{Y}^2]}\tilde{Y} = \sqrt{\frac{\mathbf{E}[\tilde{X}^2]}{\mathbf{E}[\tilde{Y}^2]}}\tilde{Y}$$

and $Y - \mathbf{E}[Y]$ is a positive multiple of $X - \mathbf{E}[X]$.